

P1355R1

Exposing a narrow contract for ceil2

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This version:

<http://wg21.link/P1355R1>

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Audience:

LEWG, LWG, SG6

Project:

ISO/IEC JTC1/SC22/WG21 14882: Programming Language — C++

Abstract

ceil2 promises an unspecified value for out-of-bounds arguments. Out-of-bounds arguments should instead be undefined behavior.

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§ 1. Background

During the review of [P0556R1], LEWG requested `ceil2` return an unspecified value for out-of-bounds arguments. [P0556R3] was adopted in Rapperswil with this specification. [N4791] ([bit.pow.two] 25.5.4 "Integral powers of 2"):

Returns: The minimal value `y` such that `ispow2(y)` is `true` and `y >= x`; if `y` is not representable as a value of type `T`, the result is an unspecified value.

At the [SAN] meeting, LEWG took a poll reconsidering this decision at the request of SG6:

Change `ceil2` to hard UB, ill-formed in constexpr

SF	F	N	A	SA
11	5	1	0	0

As discussed in [P1233R0] (out-of-bounds `shift_left` / `shift_right`), silently accepting out-of-bounds values can hide bugs. Making this explicitly undefined behavior allows it to be detected and flagged by analysis tools.

§ 2. Proposal

Wording relative to [N4791]:

```
template<class T>
    constexpr T ceil2(T x) noexcept;
```

- Let `N` be the smallest power of 2 greater than or equal to `x`.
- Expects: `N` is representable as a value of `T`.
- ~~Returns: `N` The minimal value `y` such that `ispow2(y)` is `true` and `y >= x`; if `y` is not representable as a value of type `T`, the result is an unspecified value.~~
- Constraints: ~~Remarks: This function shall not participate in overload resolution unless `T` is an unsigned integer type ([basic.fundamental]).~~
- Remarks: Call expressions that violate the precondition in the *Expects* element are not core constant expressions ([expr.const]).

§ 3. Discussion

For values of `x` that fail to satisfy the *Expects* precondition, this results in undefined behavior ([res.on.required] 15.5.4.11).

This function is *intentionally* `noexcept` with a narrow contract:

- `ceil2` is a low-level function.
- Reflector recollection of the San Diego meeting indicates this was the desire of LEWG, although minutes for this discussion are limited.

§ 4. History

§ 4.1. R0 → R1

Applied feedback from [\[PostSanDiegoReview\]](#).

- Added wording suggested by Casey Carter.
- Added clarification that having a narrow contract *and* `noexcept` is intentional.

§ References

§ Informative References

[N4791]

[Working Draft, Standard for Programming Language C++](#). 2018-12-07. URL: <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/n4791.pdf>

[P0556R1]

[P0556R1: Integral power-of-2 operations](#). 2017-03-19. URL: <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2017/p0556r1.html>

[P0556R3]

[P0556R3: Integral power-of-2 operations](#). 2018-06-06. URL: <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/p0556r3.html>

[P1233R0]

[Shift-by-negative in `shift\_left` and `shift\_right`](#). 2018-10-02. URL: <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/p1233r0.pdf>

[PostSanDiegoReview]

[LEWG\(I\) Weekly Review - P1355R0: Exposing a narrow contract for ceil2](#). 2019-01-09. URL:
<http://lists.isocpp.org/lib-ext/2019/01/9595.php>

[SAN]

[Meeting minutes for P0556](#). 2018-11-09. URL:
<http://wiki.edg.com/bin/view/Wg21sandiego2018/P0556>